Today's agenda

- Binary number system
- operators
- Problems
- Constraints

AlgoPrep

↳ decimal no. system

↳ {0-9} digits

00	10	20	90
01	11	21	91
02	12	22	92
⋮	⋮	⋮	⋮
08			
09	19	29	99

↳ Binary number system

↳ {0 1}

00	010	100	110
01	011	101	111

AlgoPrep

Conversion

↳ Convert decimal no to binary:

ex: 30

2	30	– 0
2	15	– 1
2	7	– 1
2	3	– 1
2	1	– 1
	0	

→ 11110

AlgoPrep

Quiz:

2	45	– 1
2	22	– 0
2	11	– 1
2	5	– 1
2	2	– 0
2	1	– 1
	0	

→ 101101

↳ binary no. to decimal no. → bit index

ex: $(1\ 0\ 1\ 0\ 1)_2$ → 0: unset bit / off bit
1: set bit / on bit

4	3	2	1	0
$1*2^4$	$0*2^3$	$1*2^2$	$0*2^1$	$1*2^0$

↓ add all

→ $1*2^4 + 0*2^3 + 1*2^2 + 0*2^1 + 1*2^0$

→ $16 + 0 + 4 + 0 + 1 = 21$

Quiz:

6	5	4	3	2	1	0
1	0	1	1	0	1	0

↳ $2^1 + 2^3 + 2^4 + 2^6$

$2 + 8 + 16 + 64 = 90$

→ $(1\ 0\ 1\ 0\ 2\ 0)_2$: invalid input

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4. Add binary number

$$\begin{array}{ccc} {\scriptstyle +1} & {\scriptstyle +1} & \\ 3 & 6 & 8 \\ 4 & 5 & 4 \\ \hline 8 & 2 & 2 \end{array}$$

$0+0 \rightarrow 0$

$0+1 \rightarrow 1$

$1+0 \rightarrow 1$

$1+1 \rightarrow 10$

$1+1+1 \rightarrow 11$

1

$$\begin{array}{cccccc} {\scriptstyle 1} & {\scriptstyle 1} & {\scriptstyle +1} & {\scriptstyle +1} & & \\ 0 & 1 & 0 & 1 & 1 & 1 \\ 0 & 1 & 1 & 1 & 1 & 0 \\ \hline 1 & 1 & 0 & 1 & 0 & 1 \end{array}$$

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Quiz

$$\begin{array}{ccccccc} {\scriptstyle 1} & & {\scriptstyle +1} & {\scriptstyle +1} & {\scriptstyle +1} & {\scriptstyle +1} & \\ 0 & 1 & 0 & 1 & 0 & 1 & 1 \\ 0 & 1 & 0 & 0 & 1 & 1 & 1 \\ \hline 1 & 0 & 1 & 0 & 0 & 1 & 0 \end{array}$$

Bitwise operators: { and (&), or (|), xor (^), leftShift, rightShift }

A	B	A & B (0 is dominating)	A | B (1 is dominating)	A ^ B (same same Puppy Shame)
0	0	0	0	0
0	1	0	1	1
1	0	0	1	1
1	1	1	1	0

AlgoPrep

Symbol → && | &

&& → relational operator (To Combine multiple Conditions in if)

```
if (n%2 == 0 && n%3 == 0)
```

false dominant

& → bitwise operator

|| → relational operator (true dominant)

| → bitwise operator

possible → 2 1 3 5 (binary no.) (binary no.) → binary answer ↳ decimal

ans

(i) 23 & 10 → 2

	4	3	2	1	0
23:	1	0	1	1	1
10:	0	1	0	1	0
	0	0	0	1	0

→ $2^1 = 2$

Quiz:

20:	1	0	1	0	0
10:	0	1	0	1	0
	0	0	0	0	0

→ 0

(ii) 23 | 10

	4	3	2	1	0
23:	1	0	1	1	1
10:	0	1	0	1	0
	1	1	1	1	1

$\Rightarrow 2^0 + 2^1 + 2^2 + 2^3 + 2^4 = 31$

Quiz:

	4	3	2	1	0
20:	1	0	1	0	0
10:	0	1	0	1	0
	1	1	1	1	0

$\Rightarrow 2^1 + 2^2 + 2^3 + 2^4 = 30$

(iii) 23 ^ 10

	4	3	2	1	0
23:	1	0	1	1	1
10:	0	1	0	1	0
	1	1	1	0	1

$\Rightarrow 2^0 + 2^2 + 2^3 + 2^4 = 29$

Quiz:

	4	3	2	1	0
20:	1	0	1	0	0
15:	0	1	1	1	1
	1	1	0	1	1

$\Rightarrow 2^0 + 2^1 + 2^3 + 2^4 = 27$

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Q) you have been given a Positive no., identify whether the number is even or odd.

Ex: $N = 8 \rightarrow$ even

$N = 7 \rightarrow$ odd

Note: use of +, −, *, / or % is not allowed.

$N = 10$: 1 0 1 0

+1

$N = 11$: 1 0 1 1

+1 + 0 0 0 1

$N = 12$: 1 1 0 0

+1 + 0 0 0 1

$N = 13$: 1 1 0 1

Obs: (i) Rightmost bit for even no is always 0.

(ii) Rightmost bit for odd no is always 1.

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→ To check last bit.

	3	2	1	0
N = 10 :	1	0	1	0
& 1 :	0	0	0	1
	0	0	0	0

N = 11 :	1	0	1	1
& 1 :	0	0	0	1
	0	0	0	1

N & 1 → 0 → even

N & 1 → 1 → odd

//Psuedo code

```
void checkeven (int n){
    if (n & 1 ==0) {
        s.o.p ("even");
    }
    else {
        s.o.p ("odd");
    }
}
```

Break till 9:36 Pm

// Properties

↳ (i) Commutative Property

$$a \& b = b \& a$$
$$a | b = b | a$$
$$a \wedge b = b \wedge a$$

(ii) Associative Property

$$a \& b \& c \rightarrow (a \& b) \& c = a \& (b \& c)$$
$$(a | b) | c = a | (b | c)$$
$$(a \wedge b) \wedge c = a \wedge (b \wedge c)$$

(iii) (a) $N \& N = N$

$$\begin{array}{r} N \rightarrow 110 \\ N \rightarrow 110 \\ \hline N \& N \rightarrow 110 \end{array} \qquad \begin{array}{r} N \rightarrow 110 \\ 0 \rightarrow 000 \\ \hline N \& 0 \rightarrow 000 \end{array}$$

(b) $N \& 0 = 0$

(c) $N | 0 = N$

$$\begin{array}{r} N \rightarrow 110 \\ 0 \rightarrow 000 \\ \hline N | 0 \rightarrow 110 \end{array} \qquad \begin{array}{r} N \rightarrow 110 \\ N \rightarrow 110 \\ \hline N | N \rightarrow 110 \end{array}$$

(d) $N | N = N$

(e) $N \wedge 0 = N$

$$\begin{array}{r} N \rightarrow 110 \\ 0 \rightarrow 000 \\ \hline N \wedge 0 \rightarrow 110 \end{array} \qquad \begin{array}{r} N \rightarrow 110 \\ N \rightarrow 110 \\ \hline N \wedge N \rightarrow 000 \end{array}$$

(f) $N \wedge N = 0$

Algoprep

Q) Given arr[N], every element appears twice except for one element which appears once, find that unique element.

Ex: arr[7]: { 6 8 8 7 7 10 6 } → 10

arr[5]: { 2 1 9 2 9 } → 1

// idea → Take xor of the array

arr[7]: { 6 8 8 7 7 10 6 }

1, 7 0

{ 6 ^ 6 ^ 7 ^ 7 ^ 8 ^ 8 ^ 10 } → 10

0 8 0

14 8 15 5 2

arr[7]: { ~~6~~ ^ 8 ^ ~~6~~ ^ 7 ^ 10 ^ 7 ^ 8 }

8 ^ ~~7~~ ^ 10 = 5

~~7~~

~~8~~ ^ 10 ^ ~~8~~ = 10

6 : 0 1 1 0
8 : 1 0 0 0
1 1 1 0 = 14

14 : 1 1 1 0
6 : 0 1 1 0
1 0 0 0 : 8

8 : 1 0 0 0
7 : 0 1 1 1
8^7 : 1 1 1 1 = 15

15 : 1 1 1 1
10 : 1 0 1 0
0 1 0 1 → 5

5 : 0 1 0 1
7 : 0 1 1 1
0 0 1 0 → 2

2 : 0 0 1 0
8 : 1 0 0 0
1 0 1 0 → 10

AlgoPrep

//Pseudo Code

```
int ans = 0;

for (int i=0; i<N; i++){
        ans = ans ^ arr[i];

}

s.o.p (ans);
```

T.C: $O(N)$

S.C: $O(1)$

AlgoPrep

Left Shift (<<)

	6	5	4	3	2	1	0		
a = 45 :		1	0	1	1	0	1	→	45
a << 1 :	1	0	1	1	0	1	0	→	90
a << 2 :	1	0	1	1	0	1	0	0	→ 180

↓ *2 with every left shift

$$a << N = a * 2^N$$

Qui2:

a = 1

$1 << N \rightarrow 1 * 2^N = 2^N$

↳ to calculate 2^N, use left shift operator as it is O(1) complexity operation.

AlgoPrep

Us right shift (>>)

	4	3	2	1	0		
a : 10 :	0	1	0	1	~~0~~ remove	→	10
(a>>1) :	0	0	1	0	~~1~~	→	5
(a>>2) :		0	0	1	0	→	2

divide by 2 with every right shift

$$A >> N = \frac{A}{2^N}$$

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Constraints

1 sec = 10^8 iterations

↳ array length : 10^5

↳ $O(N^2)$ → $(10^5)^2 = 10^{10}$ iterations

↳ $O(N\sqrt{N})$ → $10^5 * \sqrt{10^5} \Rightarrow 10^5 * 10^{2.5} = 10^{7.5}$ ✓

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